# Daily bike-sharing use: working versus non-working days, 2011–2012

## 1 OVERVIEW

This report compares Washington, D.C., bike-sharing use in 2011–2012, focusing on working versus non-working days. The supplied Readme defines `day.csv` as daily observations and `hour.csv` as hourly observations; only the daily file answers this question without mixing observation units. Capital Bikeshare supplied rental logs and Freemeteo weather information. Accepted numeric results are reused; new comparisons check the four day-type groups against raw records without repeating weather analyses.

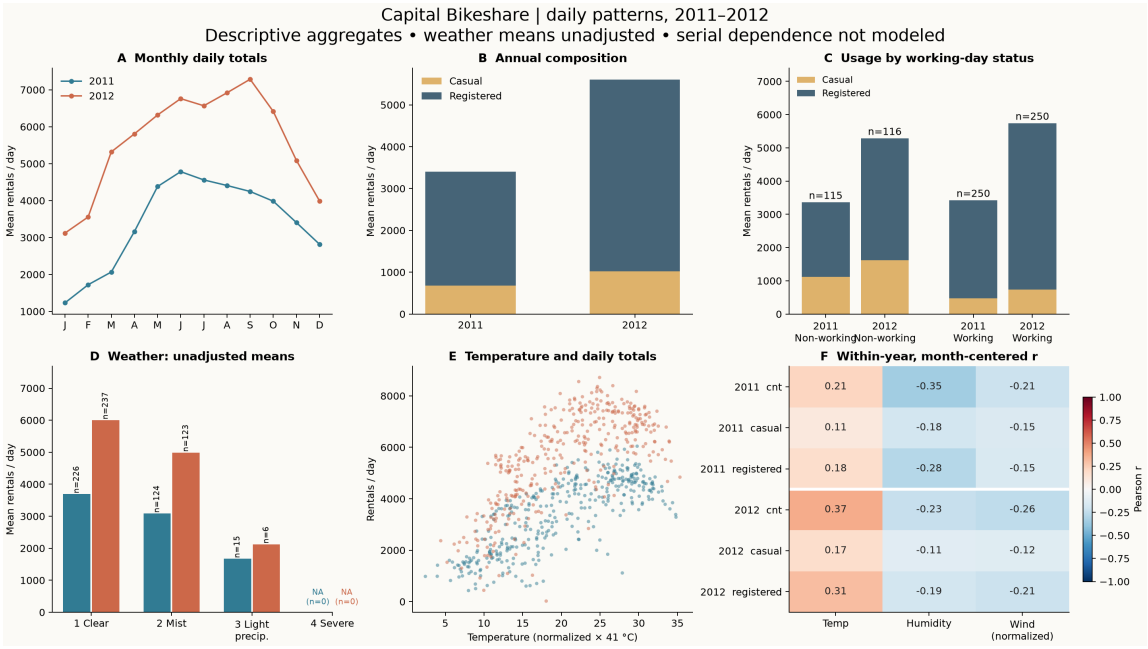



Fig. 1. Daily bike-sharing patterns across the two recorded years. (A) Mean daily totals by month. (B) Annual mean daily rentals split into casual and registered users. (C) Stacked casual/registered daily means by year and working-day status, with day counts; bar heights give total daily means. (D) Unadjusted daily totals by supplied weather category; labels show group sizes and category 4 is absent. (E) Daily total versus temperature, with the supplied normalized temperature converted by multiplying by 41 °C. (F) Pearson correlations after subtracting each year's month mean from both variables; these are descriptive month-centered associations, not causal effects or formal tests. Year colors identify 2011 and 2012 in A, D and E; component colors identify casual and registered use in B and C.

## 2 AUDIT AND METHODS

The audit reports complete coverage of all 731 dates from 2011-01-01 through 2012-12-31: 365 records in 2011 and 366 in 2012. It reports no missing dates, duplicate dates, invalid records, missing values, or nonfinite values. There were zero primary exclusions. Every row satisfied the count identity `casual+registered=cnt`. Weather-category counts were 463 for category 1, 247 for category 2, 21 for category 3, and

0 for category 4. Thus, the fourth documented weather category was absent from these daily observations rather than treated as an estimated zero-mean group.

One row, dated 2011-03-10, has `hum=0`. It is retained in all primary summaries. It is omitted only in clearly labeled sensitivity correlations, where the month means are recomputed after omission. This sensitivity rule does not assert that the value is erroneous. The audit also reports no zero-total dates.

Temperature conversion follows the supplied Readme: normalized temp × 41 °C. Humidity is normalized by 100; windspeed by 67, with no physical wind unit specified. `atemp` is audited but not used as a predictor. Daily means use every day; pooled casual share divides component sums by total sums. Weather-category means remain unadjusted.

Pearson correlations are calculated separately by year for total rentals, casual rentals, and registered rentals against normalized temperature, humidity, and windspeed. "Raw" correlations use the recorded values. "Month-centered" correlations subtract each variable's year-specific calendar-month mean before correlating the residuals. This removes monthly mean structure within each year; it is not causal adjustment.

## 3 ANNUAL USE AND COMPOSITION

Total use increased from 1,243,103 rentals in 2011 to 2,049,576 in 2012, a growth of approximately 64.9%. The daily mean increased from 3,405.8 to 5,599.9. Casual mean use rose from 677.4 to 1,018.5 rentals per day (+50.4%), while registered mean use rose from 2,728.4 to 4,581.5 (+67.9%). Consequently, registered use grew faster than casual use. The pooled casual share declined from 19.9% to 18.2%, and the mean daily casual share declined from 18.1% to 17.0%.

| Year | Days | Total rentals | Mean/day | Casual mean | Registered mean | Pooled casual share |
|---|---|---|---|---|---|---|
| 2011 | 365 | 1,243,103 | 3,405.8 | 677.4 | 2,728.4 | 19.9% |
| 2012 | 366 | 2,049,576 | 5,599.9 | 1,018.5 | 4,581.5 | 18.2% |

## 4 WORKING VERSUS NON-WORKING DAYS

Workingday is 1 for days that are neither weekends nor holidays. Non-working combines weekends and holidays. Panel C shows both component means and day counts.

| Year | Day type | Days | Total/day | Casual/day | Registered/day |
|---|---|---|---|---|---|
| 2011 | Non-working | 115 | 3,363.8 | 1,120.9 | 2,243.0 |
| 2012 | Non-working | 116 | 5,288.2 | 1,619.3 | 3,668.9 |
| 2011 | Working | 250 | 3,425.1 | 473.4 | 2,951.6 |
| 2012 | Working | 250 | 5,744.6 | 739.7 | 5,004.9 |

On non-working days, casual daily use rose 44.5%, registered use 63.6%, and total use 57.2%. On working days the increases were 56.3%, 69.6%, and 67.7%, respectively. Registered growth was larger in both groups. Working-day total means exceeded non-working means by only 61.2 rentals in 2011, but by 456.4 in 2012.

Casual means remained higher on non-working days; pooled casual shares declined from 33.3% to 30.6% there, versus 13.8% to 12.9% on working days. These do not establish travel purpose.

### 4.1 Day mix versus within-group change

Working days comprised 68.493% of 2011 and 68.306% of 2012: 250 days in each year, with one additional non-working day in 2012. Thus annual sums and daily means answer different questions. To separate mix from usage, first apply the 2011 day shares to the 2012 group means, then change to the 2012 shares. The exact daily-mean decomposition is within = sum(p2011 × (mean2012 − mean2011)); mix = sum((p2012 − p2011) × mean2012).

| Rentals/day change | Within groups | Day mix | Observed |
|---|---|---|---|
| Casual | +339.4 | +1.6 | +341.1 |
| Registered | +1,855.6 | −2.5 | +1,853.1 |
| Total | +2,195.0 | −0.9 | +2,194.2 |

At the 2011 mix, the 2012 total mean would be 5,600.8 rather than 5,599.9. Nearly all the increase therefore reflects higher means within both groups, not a different proportion of working days. This exact arithmetic uses a stated reference order; another order allocates the interaction differently. It is descriptive, not causal, and does not standardize weather or season within groups. Unrounded comparisons and decomposition are in `workingday_yoy.csv` and `day_mix_decomposition.csv`.

## 5 WEATHER PATTERNS AND ASSOCIATIONS

The supplied weather categories summarize clear/few-cloud conditions (1), mist/cloud (2), light snow/rain conditions (3), and severe weather (4). Unadjusted weather means were lower in the less favorable recorded categories. In 2011, category 1 averaged 3,695 rentals per day (n=226), category 2 averaged 3,088 (n=124), and category 3 averaged 1,674 (n=15). In 2012, the corresponding means were 6,004 (n=237), 4,991 (n=123), and 2,126 (n=6). The small category-3 samples, especially in 2012, make these descriptive contrasts sensitive to composition. Category 4 had no observations in either year.

Raw total-temperature correlations were positive: 0.771 in 2011 and 0.714 in 2012. Raw total-windspeed correlations were negative and similar across years, -0.278 and -0.279. After month centering, total-temperature correlations attenuated to 0.214 in 2011 and 0.374 in 2012; total-humidity correlations were -0.350 and -0.230, and total-windspeed correlations were -0.213 and -0.256.

| Year | Method | Total–temperature | Total–humidity | Total–windspeed |
|---|---|---|---|---|
| 2011 | Raw | 0.771 | 0.002 | -0.278 |
| 2011 | Month-centered | 0.214 | -0.350 | -0.213 |
| 2012 | Raw | 0.714 | -0.089 | -0.279 |
| 2012 | Month-centered | 0.374 | -0.230 | -0.256 |

The attenuation of the temperature association after month centering shows how much the raw association depends on monthly structure, not a causal adjustment. Omitting the zero-humidity date changes the 2011 month-centered total–humidity correlation from -0.350 to -0.385 (n=364); 2012 is unchanged (n=366). Casual and registered users both showed positive raw temperature associations, while registered use had the larger month-centered temperature association in 2012 (0.310 versus 0.173 for casual use). These patterns are consistent with differences in composition and weather-linked timing, but they should not be interpreted as effects of temperature, humidity, or wind.

No significance tests or p-values were calculated. Daily observations may be serially dependent, so they must not be treated as independent samples. The analysis estimates no causal effects and fits no predictive model.

## 6 REPRODUCIBILITY AND SOURCES

The separate revised archive retains raw copies, accepted measurements, and provenance. Run `pythonrevise.py` to rebuild only feedback tables and the revised figure; its targeted raw checks and hash checks verify reuse. `pythonanalyze.py` and `pythonverify.py` remain the initial full measurement rebuild/check workflow; then run revise.py for the final figure. Canonical `report/manuscript.md` generated the inspected LaTeX/PDF package. No unrelated analyses were rerun for this revision.

The dataset is attributed to the UCI Bike Sharing Dataset, DOI 10.24432/C5W894, with the supplied source URL at https://archive.ics.uci.edu/static/public/275/bike+sharing+dataset.zip. The required publication citation is Fanaee-T, Hadi, and Gama, Joao, "Event labeling combining ensemble detectors and background knowledge," *Progress in Artificial Intelligence* (2013), pp. 1–15, Springer Berlin Heidelberg, DOI 10.1007/s13748-013-0040-3. Field definitions and normalization conventions here follow the supplied original `Readme.txt`.